Animate Subject Tracks Workflow — User Guide

This guide walks you through loading, configuring, and running the Animate Subject Tracks Workflow, which turns an EarthRanger subject group's GPS history into an animated 3D map, and optionally an MP4 video.

What it produces

- An **interactive animated 3D map** (`animated_map.html`) with a playback bar, map buttons (zoom, compass, reset view, fullscreen), a legend of subject names, and a **Camera panel** for designing how the camera moves
- An optional **MP4 video** (`animation.mp4`) of the same animation, recorded with the same camera the map plays with. Video export is off by default.
- **GeoParquet files** of the cleaned relocations and trajectories
- A **dashboard** showing the animated map, with a **Files** section for downloading the map and the video

Requirements

- An **EarthRanger connection** configured in the workflow runner
- A **subject group** in EarthRanger, e.g. `Elephants`
- **Internet access** while the workflow runs and while you view the map, for terrain elevation and basemap imagery tiles

Longer time ranges and larger subject groups take longer to fetch and, if video export is on, to record as video. For a first run, a range of a few days to a few weeks works well.

1. Load the Workflow

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste this repository's URL into the **Github Link** field, then click **Add Template**:

```
https://github.com/wildlife-dynamics/animate-subject-tracks.git
```

Once added, the workflow appears in the **Workflow Templates** list. Click it to open the configuration form.

The card may show **Initializing...** briefly while the environment is set up.

2. Configure the Workflow

Only the **Workflow Name**, **Data Source**, **Time Range** and **Subject Group Name** need to be filled in. Every other section is pre-filled with defaults and can be left alone on a first run. Fields marked *advanced* are collapsed on the form.

Workflow Details

Field	Description
Workflow Name	A short name to identify this run, e.g. `Elephants — July 2026`
Workflow Description	Optional notes about the run

Connect to EarthRanger

Field	Description
Data Source	The EarthRanger connection to pull tracking data from

Time Range

Field	Description
Since	Start of the period to animate
Until	End of the period to animate
Timezone	Optional timezone for Since/Until
Time Format *(advanced)*	How times are displayed on the dashboard. Default `%d %b %Y %H:%M:%S`

Subject Group

Field	Description
Subject Group Name	The EarthRanger subject group to animate, picked from a drop-down of the subject groups on the chosen **Data Source**. A group with mixed subject subtypes can give unexpected results.

Trajectory Segment Filter *(advanced)*

Removes GPS noise by dropping track segments whose length, duration or speed falls outside these limits. The defaults suit most terrestrial wildlife.

Limit	Minimum	Maximum
Segment length	`0.001` m	`100 000` m
Segment duration	`1` s	`172 800` s (2 days)
Segment speed	`0.01` km/h	`500` km/h

Terrain Exaggeration *(advanced)*

Field	Default	Description
Exaggeration	`1.0`	Vertical scale of the 3D terrain. `1.0` is true scale; `2.0` doubles apparent heights, which helps on flat landscapes. The tracks are raised to match.

Terrain Sampling *(advanced)*

Track height sets where the tracks are drawn. The choices follow Google Maps 3D's altitude modes, and each shows only its own setting:

Option	Setting	What it does
Above the ground *(default)*	Height `30` m	Tracks float a set height above the ground
On the ground	—	Tracks lie on the ground; any recorded altitude is ignored

Option	Setting	What it does
Recorded altitude (where available)	Height without an altitude `30` m	Each fix at its recorded altitude above sea level. Fixes without one (usual for EarthRanger data) sit the set height above the ground instead of at sea level.
Above the 3D terrain	Height `30` m	A set height above the terrain *as drawn on the map*, so tracks, markers and labels stay on the surface at every zoom

Tracks also get an extra point every 150 m, so lines follow ridges and valleys between fixes instead of cutting through hills.

Basemap

Field	Default	Description
Basemap	Satellite	Map imagery draped over the 3D terrain, in both the interactive map and the video
Place names	Off	Adds place names and boundaries (Esri World Boundaries and Places) over the basemap

Basemap option	Description
Satellite *(default)*	Esri World Imagery aerial and satellite photos, sharp to about 0.6 m a pixel when zoomed in
Terrain	Esri World Topographic Map, with contours, roads and place names
Roadmap	Esri World Street Map
USGS Hillshade	Esri World Hillshade, a grey shaded-relief map that emphasises landforms

The basemap is given a natural colour look (slightly richer and warmer) and a soft vignette. Tracks and markers keep the colours shown in the legend.

Map Zoom & Extent *(advanced)*

The map opens centred and zoomed to fit every track. These two fields set the viewing angle, which the cameras also use.

Field	Default	Description
Pitch	`45`	Tilt in degrees (0–90). `0` looks straight down; higher values tilt towards the horizon for a 3D view.
Bearing	`0`	Rotation in degrees clockwise from north (–180 to 180). `90` puts east at the top.

Create Animation *(advanced)*

Marker icon sets what is drawn at each subject's current position:

Option	Description
Dot *(default)*	A flat circle in the subject's colour with a white outline
3D animal	A ready-tuned 3D model: elephant, giraffe, cheetah, leopard or lion, coloured by the subject's track and lit by the sun

Playback Control *(advanced)*

Field	Default	Description
Position	Bottom	Whether the playback bar sits at the bottom or top of the map
Show Camera	On	The camera button on the playback bar, which opens the Camera panel
Show map buttons	On	Zoom in/out, compass, back to the opening view, fullscreen, and a spinner while the map loads

The playback bar always has play/pause, restart, a time slider, the playback clock (e.g. `0:12 / 0:30`), the data date and time, and a speed button (0.25× to 4×).

Playback Length & Smoothness *(advanced)*

Field	Default	Description
Duration	`30` s	How long playback takes from the first location to the last. The video uses the same length by default.

Camera

Field	Default	Description
Camera settings	A fixed view of the whole scene	Settings copied from the map's **Camera panel** (or the path of its downloaded `.json` file). The interactive map plays with this camera, and so does the video.

To set a camera:

1. Run the workflow once and open the animated map.
2. Click the **camera button** on the playback bar, pick a camera and adjust it. Press play to preview it; **Use current view** fills in the centre, distance, tilt and heading from wherever you've moved the map.
3. Click **Copy settings** (or **Download**), paste them into **Camera settings**, and run the workflow again.

Camera	What it does
Fixed view	Holds one view for the whole clip; blank values frame the whole scene
Fly to & around	Flies in from the whole scene, then circles a point (blank = the centre of the data)
Follow a subject	Stays behind one subject, facing where it is heading (blank = the longest-running track)
Subject tour	Follows each subject in turn, flying between them, with each subject's name shown as the camera reaches it
Follow the action	Frames the recent movement of all subjects (or the ones you list), re-fitting as they move and spread
Cinematic fly-through	Opens on the whole scene, flies in to the action, then follows it with a tilted camera that slowly turns

Camera	What it does
Keyframes	Flies through waypoints you add in the panel, upload as a file (`.json` , `.geojson` , `.csv` or `.tsv` with lon/lat), or derive from a subject's track

The following cameras (Follow a subject, Subject tour, Follow the action, Cinematic fly-through) move smoothly and also offer **Lead room** (space ahead of moving subjects), **Zoom with the action** (in while subjects rest, out while they travel fast) and **Keep in view over hills** (the camera rises when a ridge would hide what it follows).

Video Export

Exports the animation as an MP4 video. **Video export is off by default**, so only the interactive map is produced.

Field	Default	Description
Export video	Off	Turn on to also render the animation as an MP4 video. Recording a video makes the workflow take longer to run.

Video Settings *(advanced)* only apply when **Export video** is on. The video uses the camera set in Animation → Camera.

Field	Default	Description
Duration	Auto	**Auto** matches the animation's playback length. Uncheck it to set a fixed length in seconds.
Overlays	All on	The date and time (and which corner it sits in), legend, north arrow, scale bar, title and marker labels
Resolution	720p	720p, 1080p or 4K landscape, 720p or 1080p vertical (9:16), 1080p square (1:1), or a custom width and height

3. Run the Workflow

Click **Submit**. The workflow fetches the observations, cleans them into trajectories, drapes them on the terrain and builds the animated map. If **Export video** is on, it then records the video frame by frame in a headless browser; otherwise that step is skipped.

When it's on, recording the video is usually the slowest step. The default settings give 900 frames (30 seconds at 30 fps). Higher resolutions and longer durations take proportionally longer.

4. Outputs

All files are written to the workflow's results folder (`ECOSCOPE_WORKFLOWS_RESULTS`). On ecoscope-web this is cloud storage; the video is recorded locally and uploaded there.

File	Description
`animated_map.html`	The interactive animated 3D map, shown on the dashboard as **Subject movements**
`animation.mp4`	H.264 video of the animation. Only written when **Export video** is on.
`relocations_<hash>.geoparquet`	Cleaned GPS fixes
`trajectories_<hash>.geoparquet`	Trajectory segments after the segment filter

The dashboard shows the **Subject movements** map with the workflow name, description and time range. Its **Files** section lists `animated_map.html` and, when exported, `animation.mp4` .